

# PXE Boot on UniFi — Research Findings & Implementation Log

**Date:** 2026-06-22 (session 2), original 2026-06-19 **Author:** Claude (Juniper AI) **Status:** ●  
BLOCKED — iPXE shim is CA 2011 only; CA 2011 removed from IMAGE-ME's Secure Boot db

**Immediate fix:** Disable Secure Boot on IMAGE-ME in BIOS (F1 → Security → Secure Boot → Disabled). No DHCP changes needed. Long-term fix: wait for a CA 2023-signed iPXE shim (iPXE team working on it, [rhboot/shim-review#319](#)).

## ⚠️ CURRENT BLOCKER: MS UEFI CA 2011 is expiring / removed from ThinkPad db

**Microsoft Corporation UEFI CA 2011** — the certificate used to sign ALL third-party UEFI boot applications including every Linux shim and the iPXE shim — **expires June 26, 2026** (this week).

Lenovo pushed ThinkPad firmware updates proactively removing CA 2011 from the Secure Boot db and replacing it with CA 2023. IMAGE-ME (ThinkPad P14s Gen 5) has received this firmware update.

### What's signed by what

| Component                                                          | Signing                                                                                  | IMAGE-ME result                        |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------|
| <code>ipxe-shim.efi</code> (iPXE v2.0.0)                           | MS UEFI CA 2011 <b>only</b> (1 cert in PE table)                                         | ❌ Rejected — CA 2011 not trusted       |
| <code>wimboot</code> (v2.9.0)                                      | CA 2011 + <b>CA 2023</b> (dual-signed, 2 certs in PE table)                              | ✅ Would pass Secure Boot               |
| <code>EFI/Boot/bootx64.efi</code> ( <code>bootmgfw_EX.efi</code> ) | Windows PCA 2011 (separate chain from UEFI CA)                                           | ✅ Passes (Windows chain still trusted) |
| CA 2023-signed iPXE shim                                           | <b>Does not exist yet</b> — iPXE team tracking in <a href="#">rhboot/shim-review#319</a> | —                                      |

**Key finding:** wimboot IS dual-signed and would pass IMAGE-ME's Secure Boot. But wimboot requires iPXE to inject BCD/boot.sdi/boot.wim into memory — it cannot function as a standalone PXE boot file. So the chain still needs a working iPXE shim, which requires CA 2023 signing.

## How we confirmed dual-signing

PE certificate table analysis (not Get-AuthenticodeSignature, which only reads the primary cert):

```
# on pc-deploy via WinRM:
# shimx64.efi: CertTableSize=9728  (one cert)  → CA 2011 only
# wimboot:      CertTableSize=19232 (two certs, 9728+9504) → CA 2011 + CA 2023
```

## Why no CA 2023 iPXE shim exists yet

iPXE v2.0.0 (March 6, 2026) shim was submitted to Microsoft for signing before the October 2025 dual-signing cutover, so it only got CA 2011. The iPXE team is working on a re-submission via the shim-review process (rhboot/shim-review#319). No release date. The next iPXE release will include a CA 2023-signed (or dual-signed) shim.

Source: [github.com/ipxe/ipxe/discussions/1638](https://github.com/ipxe/ipxe/discussions/1638) — iPXE collaborator NiKiZe confirmed this directly in March 2026.

## Enterprise Research: Secure Boot PXE in June 2026 (post-CA2011 expiry)

Research conducted 2026-06-23. Sources: *Lenovo Press LP2353*, *TechDirectArchive 4-part series (Parts 2/3/4)*, *miloch.dev*, *systemcenterdudes.com*, *2Pint Software*, *winreflection.com*.

## The Two Cert Chains — Critical Distinction

There are **two completely separate Microsoft Secure Boot certificate hierarchies**. Confusing them is the most common mistake in post-CA2011 PXE troubleshooting:

| Chain                                    | Signing authority                | Validates                                     | IMAGE-ME status (after Lenovo FW update) |
|------------------------------------------|----------------------------------|-----------------------------------------------|------------------------------------------|
| Microsoft Corporation<br>UEFI CA 2011    | Signs third-party UEFI boot apps | iPXE shim, Linux shims, unsigned drivers      | ✗ REMOVED from db                        |
| Microsoft Corporation<br>UEFI CA 2023    | Replacement for UEFI CA 2011     | Same, next generation                         | ✓ ADDED to db                            |
| Microsoft Windows<br>Production PCA 2011 | Signs Windows OS boot components | bootmgfw.efi, winload.efi, boot.wim internals | ✓ STILL TRUSTED                          |

**The Lenovo firmware update ONLY removed UEFI CA 2011.** The Windows Production PCA 2011 (which signs `bootmgfw_EX.efi`, `winload.efi`, and everything inside `boot.wim`) is **unaffected**.

This means: our existing `boot.wim` WinPE components (`winload.efi`, etc.) do **not** need to be replaced. They use the Windows Production PCA 2011 chain, which IMAGE-ME still trusts.

**The ONLY thing broken is the iPXE shim** — it is signed by UEFI CA 2011 (third-party UEFI CA), which is now removed from IMAGE-ME's db.

## How Enterprises Are Solving This Right Now

### Path A: WDS + Windows Server 2022 (`boot_EX` directory)

Microsoft ships CA 2023-signed WDS PXE loaders in a special directory on Windows Server 2022 only:

```
C:\Windows\System32\RemInst\boot_EX\x64\  
  wdsmgfw_EX.efi    ← CA 2023-signed WDS PXE loader  
  bootmgfw_EX.efi   ← CA 2023-signed Windows Boot Manager
```

#### Fix for WDS environments:

```
copy "C:\Windows\System32\RemInst\boot_EX\x64\wdsmgfw_EX.efi" "F:\RemoteInstall\Boot\x64\wdsmgfw.efi" /Y  
copy "C:\Windows\System32\RemInst\boot_EX\x64\bootmgfw_EX.efi" "F:\RemoteInstall\Boot\x64\bootmgfw.efi" /Y  
net stop WDSserver && net start WDSserver
```

**Caveats:** – `boot_EX` directory only exists on **Windows Server 2022** (fully patched). Does NOT exist on Windows 11. Therefore **this approach is not available for pc-deploy**. – MDT environments: when MDT rebuilds LiteTouchPE, it regenerates WinPE from the ADK baseline which still contains CA 2011 components. **MDT overwrites any manual patches** — the fix is not persistent. – As of May 2026, the ADK servicing updates (10.1.28000.1) still contain outdated WinPE binaries. True fix requires Microsoft to ship a corrected ADK. (TechDirectArchive Part 4, May 15, 2026)

**What DOES work persistently:** Build a custom WinPE outside MDT/WDS (Lenovo Press LP2353 approach). In that model, MDT never touches the WIM. For us this doesn't matter — we already build custom WinPE using ADK tools directly, without MDT.

### Path B: ConfigMgr (SCCM) 2509 PXE Responder

Starting with SCCM ConfigMgr 2509, Microsoft added a checkbox: **"Use Windows Boot Loader signed with Windows UEFI CA 2023"**

This is the **cleanest enterprise solution** for CA 2023 Secure Boot PXE. Requirements: – ConfigMgr 2509 or later – ConfigMgr PXE Responder (NOT WDS — ConfigMgr has its own PXE service) – Latest ADK + WinPE Add-on

This is a Windows Server + SCCM-licensed solution. Not applicable to our Windows 11 setup.

### Path C: iPXE + wimboot (Our Approach) — Waiting for CA 2023 Shim

Our stack (tftpd64 → iPXE shim → iPXE → wimboot → boot.wim via HTTP) is architecturally correct and actually **superior** to WDS for our use case:

| Feature                    | WDS approach                                | iPXE + wimboot approach    |
|----------------------------|---------------------------------------------|----------------------------|
| boot.wim delivery          | TFTP (requires WDS ramdisk extensions)      | HTTP (no WDS needed)       |
| Works on Windows 11 server | ❌ WDS is Windows Server-only                | ✅ Works on Windows 11      |
| MDT regeneration problem   | ❌ MDT overwrites patches                    | ✅ No MDT used              |
| Secure Boot (right now)    | ⚠️ Complex, requires boot_EX + custom WinPE | 🔄 Waiting for CA 2023 shim |
| WiFi PXE support           | ❌ TFTP too slow/large for 802.11            | ✅ HTTP scales fine         |

**The one missing piece:** a CA 2023-signed iPXE shim (UEFI CA 2023). Issue tracked: - rhboot/shim-review#319 - iPXE shim submission - github.com/ipxe/ipxe/discussions/1638 - iPXE collaborator NiKiZe confirmed CA 2023 shim in progress

Once that shim is released: drop it into `C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi`, re-enable Secure Boot on target machines. No other changes required.

### Why the 0xc0000225 Error Happens Without WDS

When using `bootmgfw.efi` directly (without iPXE/wimboot) as the TFTP boot file, Windows Boot Manager tries to download `boot.wim` via TFTP using WDS-specific TFTP ramdisk extensions. Standard TFTP servers (tftpd64, SolarWinds, etc.) do NOT implement these WDS extensions. Result: 0xc0000225.

This is confirmed independently by winreflection.com (Aug 2024) - same error, same conclusion. iPXE + wimboot bypasses this entirely by delivering boot.wim via HTTP.

### Why Our boot.wim Does NOT Need CA 2023 Updates

A common misconception (from MDT/WDS articles) is that `winload.efi` inside boot.wim needs to be updated to CA 2023. This is ONLY true in environments where the device has had the **Windows Production PCA 2011** revoked from its db - which is a separate, additional enforcement step.

IMAGE-ME (ThinkPad P14s Gen 5, Lenovo firmware update) had **UEFI CA 2011 removed** (the third-party UEFI CA), but **Windows Production PCA 2011 remains trusted**. Therefore:

- `winload.efi` inside boot.wim (signed by Windows Production PCA 2011) → ✅ still trusted
- `bootmgfw_EX.efi` (Windows Production PCA 2011) → ✅ still trusted
- Only `ipxe-shim.efi` (UEFI CA 2011) → ❌ no longer trusted

The techdirectarchive articles describe environments where **BOTH** CA 2011 chains are being revoked (some Dell SafeBIOS devices), which is a more aggressive enforcement. That situation

DOES require CA 2023 updates to WinPE internals. For IMAGE-ME, only the iPXE shim needs to change.

Sources

| Source                                   | Published  | Relevance                                                                                                          |
|------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------|
| <a href="#">Lenovo Press LP2353</a>      | 2025-2026  | Official Lenovo guide: build custom WinPE outside MDT with CA 2023                                                 |
| <a href="#">TechDirectArchive Part 2</a> | 2026-04-27 | Updating WinPE boot images with CA 2023 certs                                                                      |
| <a href="#">TechDirectArchive Part 3</a> | 2026-05-08 | Dell SafeBIOS PXE; boot_EX source files; trust gap analysis                                                        |
| <a href="#">TechDirectArchive Part 4</a> | 2026-05-15 | <b>Key:</b> ADK still broken as of May 2026; MDT overwrites patches; confirms no durable fix without MS ADK update |
| <a href="#">miloch.dev SCCM playbook</a> | 2026-05-21 | Full enterprise playbook: AvailableUpdates registry, SCCM Baselines, event-log signals                             |
| <a href="#">System Center Dudes</a>      | 2026-06-09 | SCCM boot images for hardened Secure Boot; CA 2023 checkbox in ConfigMgr 2509                                      |
| <a href="#">2Pint Software</a>           | 2024-04-19 | KB5025885 + boot WIM updates; TrustedInstaller permission issue replacing winload.efi                              |
| <a href="#">winreflection.com</a>        | 2024-08-15 | PXE without WDS on UniFi; confirms 0xc0000225 with bootmgfw.efi UEFI boot without WDS                              |

✔ DHCP OPTIONS — already applied, no changes needed

| Field                     | Current value                    | Notes                       |
|---------------------------|----------------------------------|-----------------------------|
| Boot server (Option 66)   | 192.168.5.141                    | Correct                     |
| Boot filename (Option 67) | ipxeboot/x86_64-sb/ipxe-shim.efi | Applied by Jason 2026-06-22 |
| Network Boot              | Enabled                          | Correct                     |

Recommended Interim Solution: Disable Secure Boot for Imaging

Until a CA 2023-signed iPXE shim is released:

- On IMAGE-ME: Enter BIOS (F1 at Lenovo logo) → Security → Secure Boot → Disabled → Save & Exit
- No DHCP changes needed — ipxeboot/x86\_64-sb/ipxe-shim.efi works with Secure Boot off
- PXE boot IMAGE-ME: shim → iPXE → HTTP → wimboot → WinPE → deploy.ps1

With Secure Boot disabled, UEFI skips signature validation entirely. The existing chain runs as designed. After imaging, Secure Boot can be re-enabled in BIOS; Windows boots fine either way.

**For the long-term WiFi PXE + Secure Boot goal:** the stack is ready. wimboot is already CA 2023 dual-signed. bootmgfw\_EX.efi is Windows-chain signed. When iPXE releases a CA 2023 shim, drop it into `C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi` and re-enable Secure Boot — done.

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## Architecture History (this session, 2026-06-22)

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### What happened with the bootmgfw\_EX.efi approach

A test with `ipxe-http-embedded.efi` (DHCP option 67) confirmed that the unsigned iPXE got a **Secure Boot Violation** — expected, since that binary isn't signed by any key in the ThinkPad's db.

The bootmgfw\_EX.efi approach (set earlier in the session) was superseded because: - `boot.wim` (513 MB) cannot be delivered via TFTP without WDS — UEFI TFTP has a ~4 GB limit but the Windows Boot Manager tries to load it at boot time and fails with **0xc0000225** ("Required device is inaccessible") — confirmed by WinReflection article author who hit the exact same issue. - The correct solution for non-WDS servers is: **iPXE + wimboot** (wimboot loads boot.wim via HTTP).

### Why we're using iPXE v2.0.0 (released March 6, 2026)

iPXE v2.0.0 added official Secure Boot support via a dedicated Microsoft-signed shim: - `ipxe-shim.efi` — signed by **Microsoft Corporation UEFI CA 2011** (in every PC's UEFI db) - The shim internally verifies and loads `ipxe.efi` (signed by iPXE's own CA) - No MOK enrollment required — the shim chain is fully trusted out of the box - wimboot is already signed by Microsoft — passes Secure Boot validation - `autoexec.ipxe` controls boot behavior (TFTP-fetched by iPXE, same directory as shim)

This is the correct long-term path. It also supports WiFi PXE (802.11 clients) once configured, because wimboot + UEFI HTTP avoids the TFTP size limitation entirely.

## Files deployed to pc-deploy (2026-06-22)

| Path on pc-deploy                           | Description                                      | Status         |
|---------------------------------------------|--------------------------------------------------|----------------|
| C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi | MS UEFI CA 2011-signed shim, 1,038,920 bytes     | ✔ deployed     |
| C:\tftpd64\ipxeboot\x86_64-sb\ipxe.efi      | iPXE binary (signed by iPXE CA), 397,160 bytes   | ✔ deployed     |
| C:\tftpd64\ipxeboot\x86_64-sb\shimx64.efi   | Same as ipxe-shim.efi (original name in archive) | ✔ deployed     |
| C:\tftpd64\ipxeboot\x86_64-sb\autoexec.ipxe | iPXE boot script (copy of root autoexec.ipxe)    | ✔ deployed     |
| C:\tftpd64\wimboot                          | MS-signed wimboot loader, 76,064 bytes           | ✔ pre-existing |
| C:\tftpd64\sources\boot.wim                 | WinPE (~513 MB)                                  | ✔ pre-existing |
| C:\tftpd64\EFI\Microsoft\Boot\BCD           | UEFI BCD                                         | ✔ pre-existing |
| C:\tftpd64\Boot\boot.sdi                    | Boot SDI                                         | ✔ pre-existing |

## HTTP endpoints verified (Caddy serves C:\tftpd64)

- http://192.168.5.141/ipxeboot/x86\_64-sb/ipxe-shim.efi → 200, 1,038,920 bytes ✔
- http://192.168.5.141/ipxeboot/x86\_64-sb/autoexec.ipxe → 200 ✔
- http://192.168.5.141/wimboot → 200, 76,064 bytes ✔

## autoexec.ipxe content

```
#!ipxe
echo Juniper WinPE -- Secure Boot chain (shimx64 -> ipxe -> wimboot)
set http http://192.168.5.141
kernel ${http}/wimboot
initrd ${http}/EFI/Microsoft/Boot/BCD      BCD
initrd ${http}/Boot/boot.sdi               boot.sdi
initrd ${http}/sources/boot.wim           boot.wim
boot
```

## Previous state (2026-06-22, morning session)

- CA2023 IS enrolled in IMAGE-ME's UEFI db ✔ (previous "not enrolled" diagnosis was a false negative)
- bootmgfw\_EX.efi IS properly CA2023-signed ✔

- PXECompatibility=0 is set on pc-deploy ✅
- BCD correctly configured ✅
- The "nope" failure from 2026-06-19 was almost certainly PXECompatibility=1 (TFTP log shows zero activity after the fix was applied at 3:18 PM that day)

**Remaining uncertainty (pre-iPXE pivot):** Our `bootx64.efi` is a preview/insider build (version 10.0.26100.1085, file date 4/1/2024, signing cert NotAfter 10/19/2024). A production version (3,055,456 bytes, updated 6/19/2026 by Windows Update) exists at

`C:\Windows\Boot\EFI_EX\bootmgfw_EX.efi` on IMAGE-ME. If the preview file's Authenticode hash is in the DBX, Secure Boot will still fail. This issue is moot — we've pivoted to iPXE v2.0.0 which doesn't use `bootmgfw_EX.efi`.

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## Network Topology

```
EFG Router / DHCP server (192.168.0.1)
├─ UniFi Core Switch (USW Pro 48 PoE)
│   └─ 5x UniFi U7 Pro XG APs
│       └─ Port 21 → TP-Link TL-SG108-M2 (Engineering – unmanaged)
│           ├── pc-deploy (192.168.5.141) ← Caddy HTTPS/HTTP, tftpd64 TFTP
│           ├── ThinkPad P14s Gen 5 (192.168.11.24) ← PXE client
│           └─ Yoga 7 2-in-1 (192.168.11.3) ← PXE client (WiFi only)
```

**IP space:** UniFi Default network is **192.168.0.1/20** — one flat /20 covering 192.168.0.0–192.168.15.255. All machines are on this single DHCP scope. No VLANs. No DHCP relay.

**Windows Firewall on pc-deploy: DISABLED.** All ports open.

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## Current Boot Architecture (updated 2026-06-19)

Shim has been removed. UEFI firmware now loads Windows Boot Manager directly from TFTP. This is the cleanest possible chain and is what WDS/SCCM use when Windows UEFI CA 2023 is enrolled.



```
Target UEFI firmware (Secure Boot ON)
| selects network boot entry
↓
DHCP → Option 66 = 192.168.5.141 (TFTP server)
      Option 67 = EFI/Boot/bootx64.efi (TFTP filename)
↓
TFTP: downloads EFI/Boot/bootx64.efi from 192.168.5.141:69
↓
bootx64.efi (= bootmgfw_EX.efi, Windows Boot Manager, signed by Windows UEFI CA 2023, 2,692 KB)
| verified by UEFI firmware against Windows UEFI CA 2023 in db
| IF CA2023 in db → PASSES ✓
| IF CA2023 NOT in db → Secure Boot Violation ← suspected current failure
↓
bootmgfw_EX.efi reads BCD from:
  TFTP path: EFI\Microsoft\Boot\BCD
↓
BCD: ramdisksdidevice=boot, path=\windows\system32\boot\winload.efi
     device=ramdisk=[boot]\sources\boot.wim
↓
winload.efi [⚠ currently PCA2011-signed inside boot.wim – may be a secondary issue]
↓
boot.wim (~513 MB) served via TFTP
↓
WinPE boots → deploy.ps1 → Juniper PC Deployment System
```

### ⚠ Known issue: winload.efi in boot.wim is PCA2011-signed

When boot.wim was inspected (2026-06-19): - `Windows\System32\boot\winload.efi` → **CN=Microsoft Windows Production PCA 2011**, 3,017 KB - `Windows\Boot\EFI_EX\bootmgfw_EX.efi` → **CN=Windows UEFI CA 2023** ✓ (only the boot manager has a PCA2023 variant) - No `winload_EX.efi` exists — Microsoft only provides a PCA2023 replacement for the boot manager, not the loader

KB5025885 adds **hash-based** DBX entries for specific vulnerable bootmgfw.efi versions — it does NOT broadly revoke the PCA2011 CA. Since our boot.wim's winload.efi is a different binary (ADK-built, not the vulnerable version), its hash is very likely NOT in the DBX. This should not be a problem. However, if it does fail, the fix is to rebuild boot.wim using a newer ADK (26100+) which includes updated components.

# Key Files on pc-deploy (current state as of 2026-06-19)

| Path                                    | Description                                                                  | Status           |
|-----------------------------------------|------------------------------------------------------------------------------|------------------|
| C:\tftpd64\shimx64.efi                  | Ubuntu shim, dual-signed UEFI CA 2011 + 2023, 1,048,424 bytes                | ✓ correct        |
| C:\tftpd64\grubx64.efi                  | <b>bootmgfw_EX.efi</b> (Windows UEFI CA 2023, NotAfter 2024-10-19, 2,692 KB) | ✓ <b>PCA2023</b> |
| C:\tftpd64\grubx64-ipxe-20260619.efi    | Backup of old iPXE binary (Juniper CA)                                       | 📁 backup         |
| C:\tftpd64\grubx64-pca2011-20260619.efi | Backup of PCA2011 bootmgfw.efi (NotAfter 2026-10-17)                         | 📁 backup         |
| C:\tftpd64\EFI\Boot\bootx64.efi         | Same bootmgfw_EX.efi (Windows UEFI CA 2023, NotAfter 2024-10-19)             | ✓ <b>PCA2023</b> |
| C:\tftpd64\EFI\Microsoft\Boot\BCD       | UEFI BCD: winload.efi + ramdisksdidevice=boot + block size opts              | ✓ correct        |
| C:\tftpd64\Boot\BCD                     | Legacy BCD: winload.exe + ramdisksdidevice=boot                              | ✓ correct        |
| C:\tftpd64\sources\boot.wim             | WinPE (NIC drivers injected, ~513 MB)                                        | ✓ correct        |
| C:\tftpd64\tftpd64.ini                  | PXECOMPATIBILITY=0, BaseDirectory=C:\tftpd64, Services=1                     | ✓ <b>FIXED</b>   |

## Caddyfile ( C:\caddy\Caddyfile )

Both HTTP and HTTPS serve C:\tftpd64 as the file root.

```
http://192.168.5.141 {
  log { output file C:/caddy/access.log; format json }
  file_server { root C:/tftpd64 }
}

https://192.168.5.141 {
  tls C:/caddy/server.crt C:/caddy/server.key
  log { output file C:/caddy/access.log; format json }
  file_server { root C:/tftpd64 }
}
```

# UniFi DHCP (current state — updated 2026-06-19)

| Field                   | Value                                       |
|-------------------------|---------------------------------------------|
| Network Boot            | <input checked="" type="checkbox"/> enabled |
| Network Boot — server   | 192.168.5.141                               |
| Network Boot — filename | EFI/Boot/bootx64.efi ← shim bypassed        |
| TFTP Server             | 192.168.5.141                               |

## Changes Applied 2026-06-19 (Architecture Change)

All changes made to pc-deploy via WinRM (credentials from DPAPI cache).

### 1. Replace iPXE with Windows Boot Manager (PCA2011 — later upgraded)

```
# First attempt — PCA2011-signed (deployed, then rejected by ThinkPad DBX)
Copy-Item "C:\tftpd64\grubx64.efi" "C:\tftpd64\grubx64-ipxe-20260619.efi"
Copy-Item "C:\Windows\Boot\EFI\bootmgfw.efi" "C:\tftpd64\grubx64.efi"
# Result: 2,984 KB, Issuer = Microsoft Windows Production PCA 2011 — REJECTED (in DBX)
```

### 1b. Upgrade to PCA2023-signed bootmgfw\_EX.efi

The ThinkPad has KB5025885 Stage 3: Microsoft Windows Production PCA 2011 fully revoked in DBX. The PCA2023-signed replacement ( bootmgfw\_EX.efi ) was already present in our own C:\tftpd64\sources\boot.wim at Windows\Boot\EFI\_EX\bootmgfw\_EX.efi .

```
# Mount boot.wim read-only
New-Item -ItemType Directory "C:\bootmount" -Force
dism /Mount-Wim /WimFile:C:\tftpd64\sources\boot.wim /Index:1 /MountDir:C:\bootmount /ReadOnly

# Extract PCA2023 binary
$src = "C:\bootmount\Windows\Boot\EFI_EX\bootmgfw_EX.efi"
Copy-Item "C:\tftpd64\grubx64.efi" "C:\tftpd64\grubx64-pca2011-20260619.efi" # backup
Copy-Item $src "C:\tftpd64\grubx64.efi" -Force
Copy-Item $src "C:\tftpd64\EFI\Boot\bootx64.efi" -Force

dism /Unmount-Wim /MountDir:C:\bootmount /Discard
# Result: 2,692 KB, Issuer = Windows UEFI CA 2023, NotAfter = 2024-10-19 ✓
```

Windows UEFI CA 2023 is in every modern UEFI db and is NOT in any DBX. The leaf cert's NotAfter (2024-10-19) is past, but Secure Boot validates against the CA, not the leaf cert expiry.

## 2. Fix TFTP compatibility flag

```
; tftpd64.ini (was 1, changed to 0)
PXECompatibility=0
; UEFI clients require RFC 2347 TFTP option negotiation.
; PXECompatibility=1 forces legacy mode and breaks UEFI clients.
```

## 3. Add TFTP block size optimizations to all BCD stores

```
$ramdiskGuid = "{7619dcc8-fafe-11d9-b411-000476eba25f}"
foreach ($bcd in @(
    "C:\tftpd64\Boot\BCD",
    "C:\tftpd64\EFI\Microsoft\Boot\BCD",
    "C:\tftpd64\EFI\EFI\Microsoft\Boot\BCD"
)) {
    bcdedit /store $bcd /set $ramdiskGuid ramdisktftpblocksize 16384
    bcdedit /store $bcd /set $ramdiskGuid ramdisktftpwindowsize 4
}
```

## 4. Restarted tftpd64 service

tftpd64 was restarted to pick up the `PXECompatibility=0` change.

# Why the Previous Architecture Failed

## Root cause: wrong boot chain for Secure Boot + Windows imaging

The previous chain was: `shimx64.efi` → `grubx64.efi` (iPXE signed by Juniper CA)

## Why this fails with Secure Boot ON

Shim's verification chain checks (in order): 1. UEFI `db` — list of trusted keys embedded in firmware 2. `MokList` — Machine Owner Keys enrolled per-machine via MokManager 3. Shim's embedded vendor cert — Ubuntu's Canonical cert, baked into `shimx64.efi`

The Juniper PXE CA cert is **none of these**. It's not in any UEFI `db`, it's not in `MokList` unless enrolled manually, and Ubuntu shim has no knowledge of it. Result: every boot attempt produced a `Secure Boot Violation` before iPXE could run.

## Why MOK enrollment didn't stick

Multiple attempts were made to enroll the Juniper PXE CA via MokManager (`mmx64.efi`). All failed: ThinkShield on the P14s Gen 5 appears to wipe `MokNew` between reboots when the enrollment isn't completed within the same BIOS session. A "Configuration change. Restarting." message cleared the pending enrollment each time.

Even if MOK enrollment worked, it would require manual BIOS interaction on every machine before imaging — not viable for fleet deployment.

## Why the fix works

`bootmgfw.efi` (Windows Boot Manager) is signed by `Microsoft Windows Production PCA 2011`. This certificate is **already in the UEFI db** on every PC manufactured in the last decade — it's how Windows itself boots with Secure Boot enabled. Shim loads `grubx64.efi`, verifies it against the UEFI db, finds the MS PCA 2011 cert, and passes it without any MOK enrollment.

No per-machine configuration required. This is the same chain WDS/SCCM/MDT use in every enterprise Windows deployment — the components just come from the free ADK instead of Windows Server.

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## Previous Root Cause: HTTPS URL as TFTP Filename (Resolved 2026-06-19)

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An earlier root cause was identified and fixed:

The EFG DHCP `dhcpd_boot_filename` was set to `https://192.168.5.141/shimx64.efi` (a full URL). A PXEBOOT client only speaks TFTP — it sent that literal string as the TFTP filename. `tftpd64` received an RRQ for `https:\192.168.5.141\shimx64.efi`, which is an invalid Windows path. It silently discarded the request. Zero `tftpd64` log entries. No TFTP response.

**Fix applied:** Changed to `shimx64.efi` (plain filename). PXEBOOT clients now request the correct TFTP file, which exists at `C:\tftpd64\shimx64.efi`.

**Confirmed via pktmon:** 10 TFTP RRQ packets (76 bytes each, two source ports) visible arriving at `pc-deploy:69` from ThinkPad 192.168.11.24 (MAC C8-53-09-2D-0D-56) with zero outbound response — consistent with `tftpd64` receiving an unparseable URL-format filename.

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## Testing Instructions

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### Next test (when ThinkPad is powered on)

Boot the ThinkPad P14s Gen 5 to the F12 boot menu and select the PXE/Network Boot entry (Boot0024).

**Expected success path:** 1. TFTP downloads `EFI/Boot/bootx64.efi` (= `bootmgfw_EX.efi`, PCA2023) 2. UEFI validates PCA2023 → if Windows UEFI CA 2023 in db → PASSES 3. Windows Boot Manager reads BCD → mounts `boot.wim` → WinPE loads 4. `deploy.ps1` launches — ThinkPad shows the Juniper imaging prompt

**If it still shows Secure Boot Violation:** Claude will check via WinRM whether Windows UEFI CA 2023 is enrolled in the ThinkPad's UEFI db (`Get-SecureBootUEFI -Name db`). If it's absent, the ThinkPad needs Windows Update (KB5025885) to enroll it — Claude will trigger this remotely.

ThinkPad WinRM target: 192.168.11.24 , MAC C8-53-09-2D-0D-56 , creds from lPassword.

## What success looks like

The ThinkPad displays the Windows Boot Manager loading screen (spinning dots), then WinPE loads and the deploy prompt appears.

## Secure Boot Violation — Diagnostic Log (2026-06-19)

### Attempt 1: PCA2011-signed bootmgfw.efi as grubx64.efi

Copied C:\Windows\Boot\EFI\bootmgfw.efi (PCA2011, 2,984 KB) as grubx64.efi . **Result:** Secure Boot Violation — "Invalid signature detected." **Diagnosis:** ThinkPad KB5025885 Stage 3 has hash-based DBX entries for this specific binary. ✓ confirmed.

### Attempt 2: PCA2023-signed bootmgfw\_EX.efi as grubx64.efi

Extracted bootmgfw\_EX.efi (Windows UEFI CA 2023, 2,692 KB) from boot.wim\Windows\Boot\EFI\_EX\ . Deployed as both C:\tftpd64\grubx64.efi and C:\tftpd64\EFI\Boot\bootx64.efi . DHCP option 67 was still shimx64.efi at this point — shim was loading grubx64.efi. **Result:** SAME Secure Boot Violation — "Invalid signature detected."

**Diagnosis:** Both PCA2011 and PCA2023 failing = failure is UPSTREAM of grubx64.efi, likely at shim validation level (shimx64.efi may be hash-revoked in this ThinkPad's DBX) OR Windows UEFI CA 2023 is not enrolled in the UEFI db (KB5025885 Stage 3 db enrollment hasn't run on ThinkPad).

## Change applied: Bypass shim

DHCP option 67 changed from shimx64.efi → EFI/Boot/bootx64.efi (already = PCA2023 bootmgfw\_EX.efi). The UEFI firmware now loads and validates bootmgfw\_EX.efi directly — no shim intermediary. **PENDING TEST** (ThinkPad not powered on at time of change).

## For future reference — PCA2023 source

The PCA2023 boot binary was found already inside our own boot.wim:

```
C:\tftpd64\sources\boot.wim → Windows\Boot\EFI_EX\bootmgfw_EX.efi
```

Microsoft's official script (KB5053484 / Make2023BootableMedia.ps1) copies exactly this file — there's no need to download anything external if you have a recent boot.wim.

## Full Diagnostic History

### pktmon Capture — 2026-06-19 (pre-fix)

```
pktmon start --etw -f C:\boot-capture.etl on pc-deploy during ThinkPad PXEBOOT attempt.
```

| Traffic                                  | Direction              | Finding                             |
|------------------------------------------|------------------------|-------------------------------------|
| 192.168.0.1:67 → 255.255.255.255:68      | Inbound<br>(broadcast) | EFG DHCP ACK confirmed              |
| 192.168.11.24:1753 →<br>192.168.5.141:69 | Inbound                | 5 TFTP RRQ packets (76 bytes each)  |
| 192.168.11.24:1227 →<br>192.168.5.141:69 | Inbound                | 5 more TFTP RRQ (~90 seconds later) |
| 192.168.5.141 → 192.168.11.24            | Outbound               | NOTHING — zero TFTP responses       |

10 packets, two source ports = two independent TFTP session attempts, each retried 5× on timeout. MAC `C8-53-09-2D-0D-56` = ThinkPad P14s Gen 5.

This proved: (1) EFG delivers boot options, (2) STP not blocking traffic, (3) failure on pc-deploy. Root cause: tftpd64 silently discarding RRQ with URL-format filename.

## tftpd64 Diagnostic Results — 2026-06-19

| Check                                     | Result                                                                                                                                            |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Process                                   | Running                                                          |
| <code>netstat -ano   findstr ":69"</code> | UDP 0.0.0.0:69 *.* PID — socket bound                         |
| tftpd64.log size                          | 0 bytes — no successful transfers (explained by invalid filename)                                                                                 |
| Windows Firewall                          | All three profiles OFF                                         |
| SecurityLevel                             | 0 (no IP filtering)                                            |
| tftpd32.ini vs tftpd64.ini                | Confirmed identical; tftpd64.exe reads tftpd32.ini (normal)  |
| PXECompatibility                          | Was 1 (broke UEFI TFTP option negotiation) → fixed to 0      |

## All Attempts Summary

| What                                                             | Outcome                                                    |
|------------------------------------------------------------------|------------------------------------------------------------|
| Standard PXEBOOT with URL as DHCP filename                       | TFTP arrived at pc-deploy, no response (URL filename bug)  |
| UEFI HTTP Boot with <code>http://</code> URL                     | Zero Caddy hits — ThinkPad HTTPSBOOT ignores plain HTTP    |
| UEFI HTTPS Boot (custom boot entry)                              | "Could not retrieve NBP file size" — TLS handshake fails   |
| Enroll Juniper TLS CA via BIOS cert menu                         | No change — menu is Lenovo ON-PREMISE, not UEFI HTTPS Boot |
| MOK enrollment via <code>mmx64.efi</code>                        | ThinkShield wipes MokNew; enrollment never persists        |
| Fixed URL-as-filename in UniFi DHCP                              | ✅ Fixed. TFTP now requests correct filename                |
| Fixed <code>tftpd64</code> competing DHCP (Services=15→1)        | ✅ Fixed                                                    |
| Replaced <code>grubx64.efi</code> with <code>bootmgfw.efi</code> | ✅ Applied — PENDING TEST                                   |

## Community Research Findings

| Source                      | Platform     | Finding                                                              |
|-----------------------------|--------------|----------------------------------------------------------------------|
| community.ui.com (8yr old)  | USG / EdgeOS | Garbage bytes in Option 67 — not applicable to EFG/UniFi OS          |
| community.ui.com (13yr old) | EdgeRouter   | Duplicate <code>dhcpd.conf</code> entries — not applicable           |
| kenmoini.com (2023)         | UDM-Pro      | Single-arch PXE works via UniFi DHCP; multi-arch needs external DHCP |
| HardForum (2021)            | UDM          | TFTP blocked by Windows Firewall — ruled out (WF disabled)           |
| EduGeek (Dec 2025)          | EFG          | PXE fixed by switch reboot; another user runs Windows DHCP instead   |

## References

- [Lenovo CDRT — ThinkPad Startup settings](#)
- [Lenovo CDRT — ThinkPad Network settings](#)
- [UEFI HTTPS Boot — tianocore wiki](#)
- [PXE & HTTP\(s\) Booting DHCP Options — hannan.au](#)
- [Ubiquiti Community — SCCM PXE Boot and UniFi](#)



- [pc-imaging-server repo](#)
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## Expected Boot Sequence (Secure Boot DISABLED — current interim path)

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1. IMAGE-ME PXE boots → DHCP gives option 66= `192.168.5.141` , option 67= `ipxeboot/x86_64-sb/ipxe-shim.efi`
2. UEFI TFTP downloads `ipxe-shim.efi` (1 MB) — no Secure Boot signature check
3. Shim verifies `ipxe.efi` via its own embedded iPXE vendor cert → loads `ipxe.efi` via TFTP
4. iPXE fetches `autoexec.ipxe` via TFTP from same directory
5. iPXE executes the script: downloads `wimboot` , `BCD` , `boot.sdi` , `boot.wim` via HTTP from `http://192.168.5.141/`
6. wimboot (CA 2023 dual-signed) loads WinPE from `boot.wim` in memory
7. WinPE boots → `deploy.ps1` → Juniper imaging prompt

## Expected Boot Sequence (Secure Boot ENABLED — requires CA 2023 iPXE shim, not yet released)

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1. IMAGE-ME PXE boots → DHCP gives option 66= `192.168.5.141` , option 67= `ipxeboot/x86_64-sb/ipxe-shim.efi`
2. UEFI TFTP downloads `ipxe-shim.efi` (1 MB)
3. **UEFI Secure Boot validates shim against CA 2023** → PASSES once shim is CA 2023-signed
4. Shim verifies `ipxe.efi` via embedded vendor cert → loads via TFTP
5. iPXE fetches `autoexec.ipxe` via TFTP
6. iPXE downloads `wimboot`, `BCD`, `boot.sdi`, `boot.wim` via HTTP
7. **UEFI Secure Boot validates wimboot against CA 2023** → PASSES (wimboot already dual-signed)
8. wimboot loads WinPE → `deploy.ps1` → Juniper imaging prompt

When the CA 2023 shim arrives: replace `C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi` and re-enable Secure Boot on target machines. No other changes needed.

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# Signing Evidence (collected 2026-06-22)

shimx64.efi (1,038,920 bytes, LastWrite 2026-03-06)  
PE CertTable: 9,728 bytes = ONE signature  
Cert 1: CN=Microsoft Windows UEFI Driver Publisher, issuer CN=Microsoft Corporation UEFI CA 2011  
→ CA 2011 ONLY – fails on IMAGE-ME (CA 2011 removed from Lenovo ThinkPad firmware db)

wimboot (76,064 bytes, LastWrite 2026-06-18)  
PE CertTable: 19,232 bytes = TWO signatures  
Cert 1 (9,728 bytes): CA 2011 primary  
Cert 2 (9,504 bytes): CA 2023 secondary ← confirmed by "UEFI CA 2023" string in binary  
→ DUAL-SIGNED – passes Secure Boot on IMAGE-ME 

bootx64.efi (bootmgfw\_EX.efi, 3,055,456 bytes)  
Signer: CN=Microsoft Windows, issuer CN=Microsoft Windows Production PCA 2011  
→ Windows cert chain (separate from UEFI CA chain) – IMAGE-ME trusts it 

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